EDMEALEM KASSAHUN

Software Engineer | Backend-Focused Full-Stack Developer

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PROFESSIONAL SUMMARY

Backend-focused software engineer and full-stack developer with 3+ years of experience designing and scaling web applications, backend microservices, REST APIs, and real-time systems. Strong expertise in TypeScript and Go (Golang), with production experience across Node.js, Hono, Nest.js, Express.js, Next.js, and Gin. Proficient in database architecture and schema modeling with PostgreSQL, MySQL, Drizzle ORM, GORM, and Prisma, alongside event-driven pipelines utilizing Redis and BullMQ. Proven track record taking features from architectural requirements through deployment, testing, and production support across SaaS, edtech, multi-tenant e-commerce, and execution-governance platforms.

EDUCATION

Bahir Dar University **Jun 2021 – Jun 2025**
Bachelor of Science in Software Engineering *Bahir Dar, Ethiopia*

- **GPA:** 3.86 / 4.00

PROFESSIONAL EXPERIENCE

Nexudy **Nov 2025 – Present**
Software Engineer *Seattle, WA, United States (Remote)*

- Build and maintain full-stack product features for an execution-governance platform used to align leadership, oversight, and delivery teams around verified execution.
- Develop backend services and responsive frontend workflows using TypeScript, Next.js, Node.js, Nest.js, and PostgreSQL, focusing on reliable REST APIs and clean relational data models.
- Collaborate across product and engineering to convert execution-risk and evidence-tracking requirements into production-ready software, backed by automated testing and CI/CD pipelines.

Askuala Link PLC **Mar 2025 – Oct 2025**
Backend Developer *Bahir Dar, Ethiopia (Hybrid / Remote)*

- Contributed to an AI-powered school management and communication platform serving students, parents, teachers, and school administrators.
- Implemented secure authentication and authorization flows, message broadcasting, student grading and ranking engines, annual report/roster generation, and inventory management.
- Built and maintained performant REST APIs and real-time messaging features using Node.js, Express.js, MySQL, Sequelize, and Socket.io, improving system reliability across core modules.

Nexus Tech Solutions **Nov 2024 – Feb 2025**
Full-Stack Developer Intern *Addis Ababa, Ethiopia (Remote)*

- Developed an administrative student-management dashboard for a beauty school, enabling course enrollment tracking, student payment management, and daily transaction review.
- Implemented automated payment tracking, overdue payment schedules using node-cron, SMS confirmations, and notifications via Afro Message API integration.
- Transitioned rapidly from frontend interface work into backend development, delivering reliable full-stack features with minimal supervision.

SELECTED PROJECTS

hono-commerce — Multi-Tenant E-Commerce SaaS Platform

Stack: Next.js, React.js, Hono, Bun, TypeScript, Drizzle ORM, PostgreSQL, Redis, BullMQ, Socket.io, AWS S3

- Architected an enterprise-grade multi-tenant e-commerce platform using Turborepo, Bun, Next.js (App Router), React.js, and Hono.
- Implemented 100% end-to-end type safety between the Hono backend and Next.js frontend utilizing Hono RPC and shared Zod validation schemas.
- Built an asynchronous event-driven pipeline with Redis and BullMQ featuring exponential backoff retries, dead-letter queues, and Bull-Board observability for order processing and notifications.
- Engineered dynamic wildcard subdomain routing (`[tenant].domain.com`), multi-tenant dynamic RBAC via Better-Auth, and tenant-isolated AWS S3 presigned upload workflows.

DevVault — Developer Knowledge & Second-Brain Platform

Stack: TypeScript, Next.js, React.js, Hono, Bun, Drizzle ORM, PostgreSQL, Better-Auth, Redis, BullMQ, Turborepo

- Built a modular full-stack productivity platform connecting technical notes, code snippets, linked references, and grounded AI knowledge retrieval.
- Structured a high-performance monorepo separating web frontend, Hono REST/RPC server, and background workers using Turborepo and Bun.
- Designed relational schemas using PostgreSQL and Drizzle ORM, integrating Better-Auth and a BullMQ background worker for asynchronous indexing.

Go RESTful Backend & Auth Service

Stack: Go (Golang), Gin, GORM, PostgreSQL, Docker, RS256 JWT, Viper, NATS

- Built a high-concurrency Go REST API featuring asymmetric RS256 JWT authentication (2048-bit RSA keys) and secure HttpOnly cookie delivery to prevent XSS attacks.
- Implemented modular service layers with Gin and GORM over PostgreSQL, utilizing native UUID primary keys, connection pooling, and Docker Compose orchestration.
- Structured decoupled messaging patterns and explored publish-subscribe workflows with NATS and Go concurrency primitives (goroutines, channels, sync).

Execution Governance Platform — Nexudy

Stack: Next.js, React.js, TypeScript, Node.js, Nest.js, PostgreSQL

- Developed full-stack workflows for an execution-governance platform helping institutions verify operational progress with evidence, detect risk early, and maintain a single source of truth.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Go (Golang), SQL, HTML5, CSS3

Backend & APIs: Node.js, Hono, Nest.js, Express.js, Bun, Next.js API Routes, Gin (Go), REST APIs, Hono RPC, WebSockets, Socket.io, RBAC, Authentication & Authorization (RS256 JWT, OAuth2)

Databases & Caching: PostgreSQL, MySQL, MongoDB, Drizzle ORM, GORM, Prisma, TypeORM, Sequelize, Redis, Schema Design, Relational Data Modeling

Architecture & Queues: BullMQ, NATS, Redis Pub/Sub, Event-Driven Architecture, Background Workers, Message Queues, Monorepos (Turborepo), Microservices

Frontend: React.js, Next.js (App Router), Tailwind CSS, TanStack Query, Redux Toolkit, React Hook Form, Responsive Web Design

Cloud, DevOps & Tooling: AWS (EC2, S3 Presigned URLs, RDS, SES, Route 53), Docker, Docker Compose, Linux, Git, GitHub Actions, CI/CD, Postman, Swagger/OpenAPI, Vercel, Render